SEQUENCE 1

# Arithmetic

2–11 September 2026

## 1.1 Integers, multiples and divisors

**Key point · Definitions.** $\mathbb{N} = \{0, 1, 2, \dots\}$ is the set of natural numbers. $\mathbb{Z}$ contains the natural numbers and their opposites. For $a, b \in \mathbb{Z}$, saying that $a$ is a **multiple of** $b$ means that there exists $k \in \mathbb{Z}$ such that $a = bk$. We also say that $b$ divides $a$.

$18 = 6 \times 3$: 18 is a multiple of 3 and of 6. Every integer is a multiple of 1 and of itself; 0 is a multiple of every integer. The only multiple of 0 is 0.

**Example 1.** List the positive divisors of 18 and of 24. Write three multiples of 7, one of them negative. Is 35 a multiple of 5? of 6? Justify.

**Solution.** Divisors of 18: 1, 2, 3, 6, 9, 18. Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24. For example $-7, 0, 14$ are multiples of 7. $35 = 5 \times 7$ is a multiple of 5; $35 = 6 \times 5 + 5$ is not a multiple of 6.

**Key point · Sum of multiples.** The sum of two multiples of an integer $a$ is a multiple of $a$.

**Proof to write.**

**Solution.** Write $u = ak$ and $v = a\ell$, with $k, \ell \in \mathbb{Z}$. Then $u + v = a(k + \ell)$ and $k + \ell \in \mathbb{Z}$. So $u + v$ is a multiple of $a$. This also holds for $a = 0$.

**Fluency · 5–10 min.** Calculate $\frac{2}{3} + \frac{1}{6}$; $3(x + 2) - x$; the image of 2 under $x \mapsto 2x - 3$.

**Answer.** $\frac{5}{6}$; $2x + 6$; 1.

## 1.2 Parity and prime numbers

**Key point · Definitions.** An integer $n$ is **even** if it can be written $2k$, with $k \in \mathbb{Z}$, and **odd** if it can be written $2k + 1$. A **prime number** is a natural number greater than or equal to 2 that has exactly two positive divisors: 1 and itself.

The prime numbers less than 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

**Key point · Property to prove.** The square of an odd integer is odd.

**Proof to write.**

**Solution.** If $n = 2k+1$, then $n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$. Since $2k^2+2k \in \mathbb{Z}$, $n^2$ is odd. The final form justifies the parity; a few numerical trials are not enough to prove the statement.

**Method · Testing whether an integer is prime.** For $n \geq 2$, it is enough to test the prime divisors less than or equal to $\sqrt{n}$. Indeed, if $n = ab$ with $1 < a \leq b$, then $a \leq \sqrt{n}$; $a$ has a prime divisor which also divides $n$.

**Example 2.** Show that 157 is prime and that 161 is not.

**Solution.** Since $12^2 < 157 < 13^2$, we test 2, 3, 5, 7, 11. None divides 157: for 7, $157 = 7 \times 22 + 3$; for 11, $157 = 11 \times 14 + 3$. So 157 is prime. On the other hand, $161 = 7 \times 23$ is composite.

## 1.3 GCD and irreducible fractions

**Key point · Definition and convention.** For two integers $a, b$ **not both zero**, $\gcd(a, b)$ is the greatest of their common positive divisors. We have $\gcd(a, b) = \gcd(|a|, |b|)$ and $\gcd(a, 0) = |a|$ if $a \neq 0$. By convention, we set $\gcd(0, 0) = 0$; this last case does not come under the definition by the greatest divisor.

For example, $\gcd(18, 24) = 6$, and changing the signs of 18 or 24 does not change this result.

**Key point · Prime factorisation – accepted.** Every natural number greater than or equal to 2 is a product of prime numbers. This factorisation is unique up to the order of the factors.

**Example 3.** Factorise 140, 150, 630 and 120 into prime factors.

**Solution.** $140 = 2^2 \times 5 \times 7$; $150 = 2 \times 3 \times 5^2$; $630 = 2 \times 3^2 \times 5 \times 7$; $120 = 2^3 \times 3 \times 5$. We divide successively by prime numbers and check the final product.

**Method · Making a fraction irreducible.** For $b \neq 0$, divide the numerator $a$ and the denominator $b$ by $\gcd(a, b)$. The fraction obtained has a numerator and a denominator that are coprime.

**Example 4.** Make $\frac{140}{630}$, $\frac{84}{126}$ and $\frac{-120}{150}$ irreducible.

**Solution.** The GCDs are 70, 42, 30 respectively. Thus $140/630 = 2/9$, $84/126 = 2/3$ and $-120/150 = -4/5$. For 140/630, we can also cancel the factors 2, 5, 7 in the factorisations.

## 1.4 Set vocabulary in context

**Key point · Definitions and notation.** $x \in A$ means that $x$ is an element of $A$; $A \subseteq B$ means that every element of $A$ belongs to $B$. $\varnothing$ is the empty set.

- $A \cap B$: elements belonging to $A$ **and** to $B$;
- $A \cup B$: elements belonging to $A$ **or** to $B$ (possibly both);
- in a reference set $E$, $\overline{A} = E \setminus A$ contains the elements of $E$ that are not in $A$;
- for a finite set, $\mathrm{Card}(A)$ is its number of elements.

**Example 5.** In $E = \{1, 2, 3, 4, 5, 6\}$, let $A = \{2, 4, 6\}$ and $B = \{3, 6\}$. Give $A \cap B$, $A \cup B$, $\overline{A}$ and $\mathrm{Card}(A)$. Are the statements $2 \in A$, $\{2\} \subseteq A$ and $A \subseteq B$ true?

**Solution.** $A \cap B = \{6\}$, $A \cup B = \{2, 3, 4, 6\}$, $\overline{A} = \{1, 3, 5\}$, $\mathrm{Card}(A) = 3$. The first two statements are true. The third is false, for example $2 \in A$ and $2 \notin B$.

**Key point · Pair and Cartesian product.** A pair $(a, b)$ is ordered: in general $(a, b) \neq (b, a)$. The Cartesian product $A \times B$ is the set of pairs $(a, b)$ with $a \in A$ and $b \in B$. For finite sets, $\mathrm{Card}(A \times B) = \mathrm{Card}(A)\,\mathrm{Card}(B)$.

**Example 6.** Write all the elements of $\{1, 2\} \times \{3, 4, 5\}$.

**Solution.** $\{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$. There are $2 \times 3 = 6$ pairs.

## 1.5 Algorithms and putting it to use

**Method · Python: integer, Boolean and condition.** `a % b` is the remainder of the Euclidean division of `a` by `b` for every integer `a` when $b > 0$, the case used here (if $b < 0$, Python returns a result with the sign of $b$, which is not the Euclidean remainder). A test returns a Boolean: `True` or `False`. Equality is tested with `==`; assignment uses `=`.

```
def est_multiple(a, b):
    if b == 0:
        return a == 0
    return a % b == 0
```

**Example 7.** Predict the results of `est_multiple(35, 5)` and `est_multiple(35, 6)`, then of `est_multiple(0, 7)`, `est_multiple(7, 0)` and `est_multiple(0, 0)`. Explain the role of the condition.

**Solution.** The results are `True`, `False`, `True`, `False`, `True`. The case $b = 0$ avoids calculating a remainder modulo zero, which is impossible, and applies the definition: only 0 is a multiple of 0.

**Example 8 · Problem.** We have 84 notebooks and 126 pencils. We want to make as many identical packs as possible, using all the material. How many packs can we make? What does each contain?

**Solution.** The number of packs divides 84 and 126. Its maximum is therefore $\gcd(84, 126) = 42$. Each pack contains $84/42 = 2$ notebooks and $126/42 = 3$ pencils. The check $42 \times 2 = 84$, $42 \times 3 = 126$ confirms that all the material is used.

**Teacher note.** Diagnostic: have students state “divides” and “is a multiple of”. Help: start by testing 2, 3, 6 packs. Extension: prove that $n(n+1)$ is even by distinguishing the two parities of $n$.